

OLIVERB MK II

*A valve preamp, a passive high-pass filter, a two-track tape echo, and a spring tank
— the boxes that made dub, in one plug-in.*

USER MANUAL · V2.0.0 · VST3 / AU / STANDALONE · MACOS ·
WINDOWS · LINUX

1 THE INTERFACE AT A GLANCE

OLIVERB is one dark faceplate with five color-striped sections, stacked top to bottom in the order the signal flows (in the default routing). Every knob shows its value beneath it; every section has its own on/off switch on the left. **New in MK II:** the amber **Valve** section — a triode preamp for real even-harmonic tube saturation.



Figure 1 — The OLIVERB MK II faceplate at its default settings. The colored stripes on the left edge of each section match the headings used through this manual.

SECTION	WHAT LIVES THERE
 Header	Preset selector, Bypass , and the master Output trim (−24 to +12 dB).
 Valve — Preamp	Drive, Bias, Sag, Tone and Mix, the filament glow lamp, and three switches: Valve on/off, Output (moves it to the end of the chain) and Echo Amp (the same valve inside the tape machine).
 Filter — Big Dial	The red frequency dial with its eleven silkscreened positions, six character knobs (Impedance, Magnetism, Character, Dynamics, Artefacts, Gain), and three switches: Filter on/off, Bank B , and Post .
 Echo — Two Track	Time, Feedback, Input, Output, Hiss, Mix and Wear, plus Echo on/off, Sync (tempo lock) and the Dub Send switch.
 Mod — LFO / Envelope	Rate, LFO Depth, Env Depth, Sens and Speed, the LFO shape selector, and the LFO / Sync switches. Both sources drive the big dial.
 Spring — Tank	Spring (amount), Tension (decay) and Drive, with the Spring on/off switch.

2 INSTALLATION

MACOS

Download `OLIVERB-macOS.zip` from the [releases page](#), unzip it, and run **OLIVERB-macOS.pkg**. It installs the VST3 and AU for all users into `/Library/Audio/Plug-Ins/VST3` and `/Library/Audio/Plug-Ins/Components`.

The installer is unsigned, so the first launch may be blocked: right-click the `.pkg` → **Open** → **Open** anyway. If the plug-in itself is quarantined:

```
xattr -dr com.apple.quarantine /Library/Audio/Plug-Ins/VST3/OLIVERB.vst3
```

Prefer a manual install? The zip also contains the raw `OLIVERB.vst3` and `OLIVERB.component` — drop them into `~/Library/Audio/Plug-Ins/VST3/` and `~/Library/Audio/Plug-Ins/Components/`.

WINDOWS

Download `OLIVERB-Windows.zip`, unzip, and run **OLIVERB-Windows-Setup.exe**. It installs to `C:\Program Files\Common Files\VST3\OLIVERB.vst3`. SmartScreen will warn about an unrecognised app: click **More info** → **Run anyway**. For a manual install, copy the `OLIVERB.vst3` folder into `C:\Program Files\Common Files\VST3\`.

LINUX

Download `OLIVERB-Linux.zip` and copy `OLIVERB.vst3` to `~/vst3/`.

After installing, rescan plug-ins in your DAW. OLIVERB appears under **Deestech** in the Fx / Filter / Delay / Reverb categories. Upgrading from v1: install over the top — sessions saved with v1 load with the valve at its default settings.

3 FIRST SOUND IN SIXTY SECONDS

1. Insert OLIVERB on a drum loop or a full mix.
2. Open the preset menu and pick **Waterhouse Rockers** — the classic chain: valve into filter into echo into spring.

3. Turn the valve **Drive** up and watch the glow lamp: the sound thickens and compresses but doesn't get louder. That's the tube.
4. Grab the big **Frequency** dial and sweep it up and back down. The corner peak, the switch clacks, the way the echo tail follows the dial — that's the plug-in.
5. Turn **Feedback** past 80 % and tap **Dub Send**: the input stops feeding the machine but the loop keeps regenerating. That's a dub-out.
6. Tap it back and carry on.

4 SIGNAL FLOW

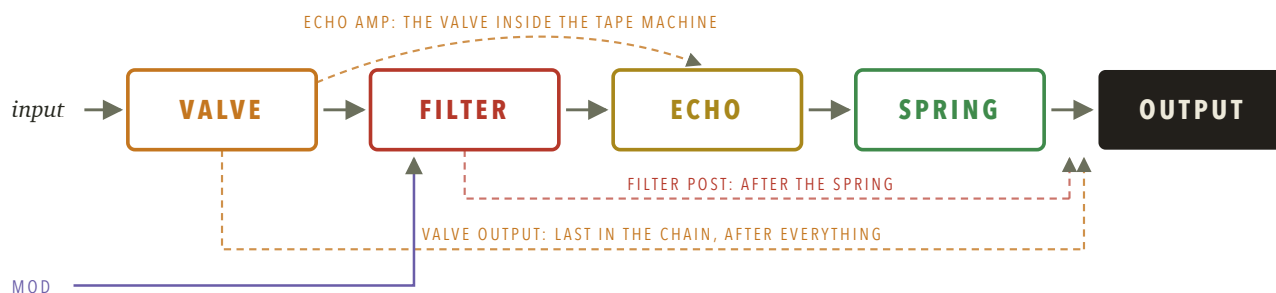


Figure 2 — The signal chain. The valve sits first by default or last with Output lit; Echo Amp puts the same valve inside the tape machine. The Mod section steers the filter's frequency dial wherever the filter sits.

The **Valve** colours everything after it at the front, or becomes a master valve stage across the whole mix, echo tails and spring included, at the back. The filter's **Pre** / **Post** switch is the other big routing decision:

- **Pre** — the filter feeds the echo, so every repeat inherits the dial position, and sweeping the dial drags the echo tail with it. This is the performance position.
- **Post** — the filter sits across the finished signal, echo tails included. Use it to carve the whole wet mix at once.

Everything runs at 2× oversampling internally; the extra latency is reported to your host automatically, so tracks stay aligned.

5 THE VALVE – PREAMP

A single-ended triode line amplifier with an output transformer. Its transfer curve is asymmetric — grid conduction flattens the positive swing quickly, cut-off softens the negative swing slowly — which is why it makes even harmonics (warmth) rather than the odd-only fuzz of a symmetric clipper. The **glow** lamp shows how hard the valve is actually working.

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Drive	0–100 % (35 %)	Gain into the grid, up to +36 dB. Level is compensated on the way out, so Drive changes the <i>shape</i> of the sound far more than its loudness. Zero is a clean line amp (about 1 % second harmonic); full is a valve on its knees.
Bias	0–100 % (50 %)	The operating point. Cold is near-symmetric and stays clean until pushed; hot is asymmetric from the first volt and thick at any level — the even-harmonic knob.
Sag	0–100 % (30 %)	The supply giving way under load: gain drops and bias goes colder as the signal rises. Compression, bloom, a stage that pushes back.
Tone	0–100 % (40 %)	The output transformer: iron bump below 100 Hz, loss at the very top (19 kHz down to 6.5 kHz), and the core rounding off what's still too big for it.
Mix	0–100 % (100 %)	Parallel blend against the dry signal.
Output	Input / Output (Input)	<i>Input</i> puts the valve first in the chain; <i>Output</i> puts it last, after the spring.
Echo Amp	(On)	Hands the same valve curve to the tape machine's record amplifier, scaled by Drive. Off restores the original solid-state record amp.

6 THE FILTER – BIG DIAL

A passive, inductor-based high-pass, third order (18 dB/octave). No active parts — its character comes from termination and core saturation, not resonance.

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Frequency	1–11 (1)	Eleven switch positions — see the table below.
Bank B	A / B (A)	Capacitor bank. A is the classic broad sweep; B sits lower and tighter.
Impedance	0–100 % (35 %)	Termination. At 0 % the network is properly terminated and flat; open it up and the corner lifts by several dB. This peak is the sound of the hardware.
Magnetism	0–100 % (30 %)	How hard the inductor cores saturate. Signal level pushes the corner upward and generates harmonics.
Character	0–100 % (25 %)	Non-linearity in the damping path — harmonic content, independent of the core.
Dynamics	0–100 % (30 %)	How fast the core tracks the signal. Slow = a gentle breathing lift; fast = it behaves like an envelope filter.
Artefacts	0–100 % (25 %)	Switch thump. At 0 % the corner glides between positions (the hardware can't). At 100 % it clacks like the real rotary.
Gain	±18 dB (0)	Internal trim after the filter.
Post	(off = Pre)	Filter before the echo, or across the finished signal — see §4.

SWITCH POSITIONS

STEP	1	2	3	4	5	6	7	8	9	10	11
Bank A	70	100	150	220	330	500	750	1.1k	1.8k	3.5k	7.5k
Bank B	50	80	120	170	240	350	520	800	1.3k	2.4k	5.0k

All values in Hz.

7 THE ECHO – TWO TRACK

A studio tape recorder used as an echo: record head → tape → replay head, output patched back to the input. Each pass through the loop loses a little top end and gains a little saturation, so long tails get darker and thicker rather than louder. With the valve's **Echo Amp** lit, that saturation is the triode curve.

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Time	20–2000 ms (375)	Tape delay. Changing it drags the transport, so pitch bends on the way — as tape does.
Sync	(Free)	Lock Time to the host tempo. When lit, Time becomes a note division, 1/16 to 1/1 with triplet and dotted values (default 1/4).
Feedback	0–100 % (34 %)	Loop regeneration. Unity sits at 80 % — above that the machine self-oscillates on purpose, limiting into the record amp instead of exploding.
Dub Send	Dub / Held (Dub)	<i>Dub</i> feeds the input into the machine. <i>Held</i> closes the door: the input stops, but whatever is on the loop keeps circulating.
Input	0–100 % (57 %)	Level into the record amp. Push it to saturate the tape harder.
Output	0–100 % (60 %)	Level out of the machine.
Hiss	0–100 % (18 %)	Tape noise, recorded <i>to</i> the tape — it recirculates and builds with feedback instead of sitting on top.
Mix	0–100 % (32 %)	Dry/wet balance for the echo.
Wear	0–100 % (45 %)	Machine condition: one knob scaling flutter, HF loss, head bump and record-amp drive from freshly-aligned to tired.

8 MOD – LFO / ENVELOPE

Both modulation sources push the **Frequency dial**, in octaves relative to the switch position, so everything the filter does — corner peak, core saturation, character — rides along with the sweep. Modulation is shared by both channels, so the stereo image holds.

LFO

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
LFO	(Off)	Enables the LFO.
Shape	5 shapes (Sine)	Sine, Triangle, Saw Down, Square, S+H. Square and S+H are lightly smoothed: the corner snaps, the zipper doesn't.
Rate	0.02–20 Hz (0.8)	Free-running speed.
Sync	(Free)	Lock the rate to the host tempo, 1/16 to 1/1 (default 1/2); synced sweeps re-align to the bar.
LFO Depth	±3 oct (0)	How far the dial travels. Negative inverts the sweep.

ENVELOPE FOLLOWER

Follows the **input** signal (pre-filter), so it responds to what you play, not to what the filter is already doing.

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Env Depth	± 3 oct (0)	Bipolar: positive opens the filter on hits (auto-wah); negative ducks it out of the way.
Sens	0–100 % (50 %)	How hard the follower listens.
Speed	0–100 % (50 %)	How fast it moves.

9 THE SPRING – TANK

A two-tank spring reverb. Springs are dispersive — highs travel through the coil faster than lows — so a transient smears into the characteristic descending *boing* that room reverbs never produce.

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Spring	0–100 % (22 %)	How much of the tank is in the output.
Tension	0–100 % (55 %)	Decay time of the tank.
Drive	0–100 % (25 %)	Level into the send transducer. This is where the crash lives — hit it hard and transients splash.

10 GLOBAL

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Output	–24 to +12 dB (0)	Final level trim.
Bypass	(Off)	True bypass of the whole plug-in.

Each of the four sections also has its own **On** switch, so OLIVERB can serve as just a valve, just a filter, just an echo, or just a spring.

11 FACTORY PRESETS

PRESET	WHAT IT SHOWS
Init	Everything at defaults — the valve on at a gentle 35 %.
Waterhouse Rockers	The classic chain at working settings, valve warm — the place to start.
Valve Warmth	Valve only, everything else off: a tube line amp on a bus or a mix.
Hot Preamp	Valve only, driven hard with heavy sag: a mic pre on its knees.
Snare Throw	High feedback, hot input: hit it with one snare and ride the tail.
Big Dial Sweep	Open termination, heavy magnetism, audible switch clacks — for performing the dial.
Tape Wash	Long, worn, hissy repeats that melt into the spring, valve tone thick.
Glowing Tape	The valve at the <i>output</i> , driven, with a hot tape loop — everything cooks together.
Held Echo (Dub Out)	Send is <i>Held</i> and feedback at unity: an infinite loop, filter across the output. Open the Send to let new signal in.
Spring Crash	The tank up front, driven hard.
Roots Bass Tighten	Filter and valve only, step 1, a little core saturation — a bass-tightening tool, no wet signal at all.
Auto Wah Skank	Envelope follower opening the dial on hits. Guitars and claws.
Tidal Sweep	Tempo-synced triangle LFO sweeping two octaves over a whole bar.
Siren	Everything at maximum, valve included. You were warned.

12 TECHNIQUES

- **Valve as a bus tool.** Load *Valve Warmth*. Drive 30–50 %, Bias 60 %, Sag 20 %: glue and weight without obvious distortion. The glow should flicker, not burn.
- **Pushed preamp.** Drive past 70 % with Sag up: the stage compresses into itself and blooms after transients. Bias hot for thickness, cold for a cleaner crunch.
- **Valve at the back.** Flip *Output* so the valve sits after the spring: echo tails and splashes get rounded off by the same tube, and hot feedback stops sounding digital.
- **The dub throw.** Feedback high, Mix high, Send on *Dub*. Un-mute one hit — a snare, a vocal word — then flip Send to *Held*. The hit circulates and decays on its own while the dry signal carries on. Flip back to *Dub* for the next throw.
- **Ride the dial.** With the filter **Pre**, sweep Frequency during the echo tail: every repeat re-inherits the new corner and the whole tail bends. Add **Artefacts** for the hardware clack on each step.
- **Pitch-bend echoes.** Automate **Time**. The transport drags, so repeats bend like varispeed tape. Small moves = subtle warble; big jumps = dive-bombs.
- **Self-oscillation instrument.** Feedback past 80 %, no input: the machine sings on its own. Tune it with Time, filter it with the dial, crash it into the spring.
- **Auto-wah.** Env Depth positive, filter step 4–6, Impedance up. Sens sets how hard it listens, Speed how fast the corner chases the playing.
- **Reverse duck.** Env Depth *negative*: the filter closes on hits and blooms back open in the gaps — an ungate for pads and textures.

13 TROUBLESHOOTING

SYMPTOM	FIX
Plug-in doesn't appear in the DAW	Rescan plug-ins. Confirm the <code>.vst3</code> is in the system VST3 folder (§2). On Apple-silicon Macs check the DAW isn't running in Rosetta with an arm64-only scan cache.
macOS blocks the installer or plug-in	Right-click → Open on the <code>.pkg</code> , or clear quarantine with the <code>xattr</code> command in §2.
Windows SmartScreen warning	More info → Run anyway. The installer is unsigned, not unsafe.
Sound is late or flaming	OLIVERB reports its $2\times$ oversampling latency to the host; enable your DAW's plug-in delay compensation.
Everything sounds slightly thicker than v1	That's the valve at its default 35 %. Turn the Valve switch off, or Drive to 0, for the v1 sound.
Echo tail never dies	Feedback is at or above 80 % (unity). That's a feature — pull it down, or flip Send to <i>Dub</i> with no input.
Output slammed after big dial moves	Open Impedance plus hot echo feedback genuinely adds level. Use the filter Gain or global Output trim. The output stage is hard-limited, so it cannot run away.

14 SPECIFICATIONS

- **Valve:** single-ended triode model, unity-normalised asymmetric transfer, +36 dB drive range, level-compensated, supply sag, transformer voicing
 - **Formats:** VST3, Audio Unit (macOS), Standalone application
 - **Platforms:** macOS 10.13+ (universal: Apple silicon + Intel), Windows 10+ (64-bit), Linux
 - **Processing:** 32-bit float, $2\times$ oversampled (half-band polyphase IIR), latency-compensated
 - **Filter:** third-order (18 dB/oct) passive constant-k model, TPT state-variable core, stable under audio-rate modulation
 - **Echo:** 20–2000 ms, wow & flutter, per-pass HF loss, in-loop record-amp saturation (solid-state or valve)
 - **Spring:** two dispersive tanks (33.7 / 41.9 ms coil transit), twelve allpass sections each
 - **Source & license:** MIT, at github.com/damielolar-CODE/oliverb
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